

Matthew Ryan Jones

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EDUCATION

Drexel University - Philadelphia, PA

Bachelor of Science in Biomedical Engineering | GPA: 3.4

Concentration: Bioinformatics | Minor: Mathematics

Relevant Coursework: Scientific Data Analysis I & II, Biomedical Statistics, Experimental Design in Biomedical Research, Numerical Analysis, Computational Bioengineering, Biomedical Systems and Signals, Brain-Computer Interface Laboratory

WORK EXPERIENCE

Mayo Clinic – Data Science Intern

October 2023 – March 2024

- Developed a tissue capture algorithm using machine learning for breast cancer pathology slides, increasing diagnostic accuracy in pathology workflows.
- Filtered digital slides through a pre-trained Convolutional Neural Network (CNN, ResNet) to extract feature data supporting downstream classification models.
- Researched quality control improvements for whole-slide image (WSI) artifact detection, enhancing reproducibility in automated analysis.
- Integrated PathProfiler pipeline from GitHub to perform automated quality assessment on 100+ WSIs, identifying artifacts and usability issues to accelerate slide review.
- Leveraged the open-source Digital Slide Archive platform to visualize and manage WSIs, supporting quality review and feature validation.
- Co-authored an abstract submission to *Modern Pathology* (USCAP), contributing to peer-reviewed scientific research.

Ethos Biosciences – Process Engineer Co-op

September 2022 – March 2023

- Devised and executed validation protocols (IQ/OQ/PQ) for 15+ in-house assets, achieving ISO-13485 compliance and audit readiness.
- Created Standard Operating Procedures (SOPs) for 25+ manufacturing and warehouse operations, standardizing procedures across departments for employee training purposes.
- Managed documentation change requests for Quality Assurance/Quality Control, working cross-functionally to maintain cGMP compliance and reduce approval turnaround times.
- Compiled a comprehensive equipment tracking database in Excel, documenting validation dates, IDs, and status criteria to support facility audits and ongoing asset management.
- Performed preventative maintenance checks, improving equipment reliability and workspace safety.

Drexel University – Academic Center for Engineers Tutor

May 2024 – September 2024

- Educated first- and second-year engineering students, improving exam performance through tailored one-on-one instruction.
- Instructed students in thermodynamics and physics, reinforcing critical thinking and analytical problem-solving skills.
- Prepared customized practice material prior to sessions, optimizing study efficiency and strengthening comprehension of key concepts.

ENGINEERING PROJECTS

Bioinformatics Course Project – Differentially Expressed Gene (DEG) Identification in Patients with Heart Disease

- Analyzed public datasets using Principal Component Analysis (PCA) and hierarchical clustering, evaluating sample separation and improving data interpretability in a visually appealing format.
- Identified 80+ differentially expressed genes based on fold-change and statistical p-values, highlighting biomarkers of heart disease.
- Enriched pathways to reveal 16 significant gene ontology terms with open-source DAVID, linking DEGs to biological functions.
- Trained and cross-validated a binary Support Vector Machine (SVM) classifier, achieving accurate prediction of healthy vs. diseased patients from gene expression data.

Senior Design Capstone – An Optogenetics Device for Engineering Neural Circuits in *Drosophila melanogaster*

- Consulted with the senior advisor and PhD candidate to define user requirements and establish constraints, guiding the 30-week design process.
- Designed an Arduino-based temperature control system, maintaining specimens in the pupa stage - a critical condition for neurological scanning of *Drosophila* in neurodegenerative disease studies.
- Developed two test-case C++ scripts, verifying sensor accuracy and ensuring fan activation at correct thresholds.
- Collaborated with teammates and advisors in weekly progress updates, integrating feedback to refine system performance.
- Presented design specifications and solution overview to engineering faculty, effectively communicating technical concepts to a professional audience.

TECHNICAL SKILLS

Programming & Tools: MATLAB (Simulink), R, Julia, SQL, C++, Python, GitHub, Visual Studio Code, Microsoft Excel and Visio, Unix/Linux Operating Systems

Data Analysis & Modeling: Data collection, hypothesis testing (parametric & non-parametric tests), statistical modeling, machine learning (Convolutional Neural Networks/CNNs, Support Vector Machine/SVM, Linear Discriminant Analysis/LDA), PCA, clustering

Data Processing & Visualization: Normalization techniques, data querying and transformation (SQL, dplyr), visualization (ggplot2, Excel), reporting, Microsoft Office Suite

Engineering & Hardware: Arduino microcontrollers, C++ scripting, electrical schematics, sensor validation, temperature control systems

Languages: Spanish (Advanced) - NJ State Seal of Biliteracy; Excellence in Spanish Language Award 2020